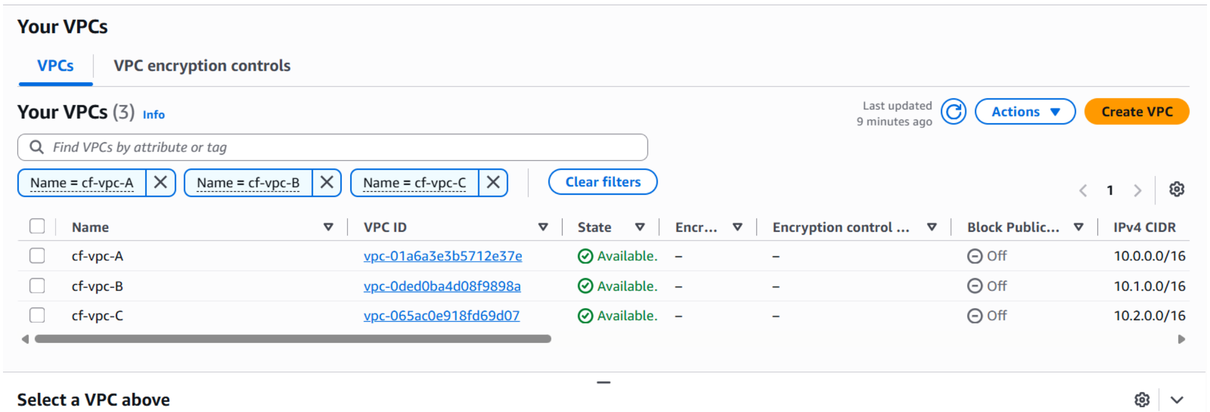
Configuring an AWS Transit Gateway

To Begin with the Lab

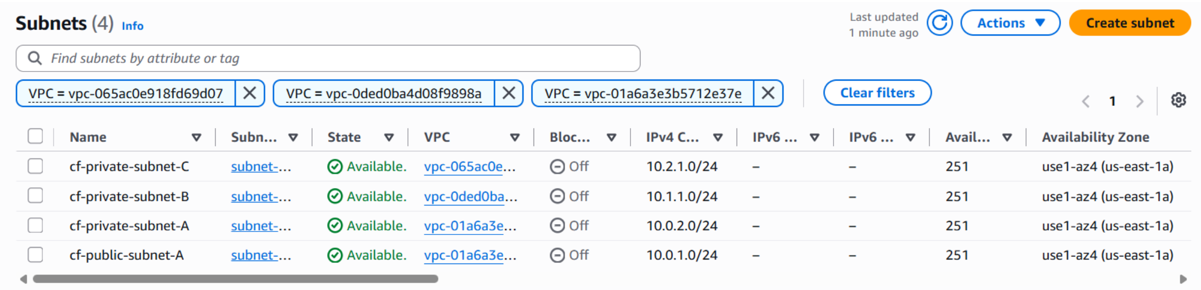
Summary of the Lab

Note: This lab assumes that three VPCs have already been created. **VPC-A** contains one public subnet (for the Jump Host) and one private subnet, while **VPC-B** and **VPC-C** each contain one private subnet. The EC2 instances and security groups are already configured. The objective of this lab is to connect all three VPCs using an **AWS Transit Gateway**.

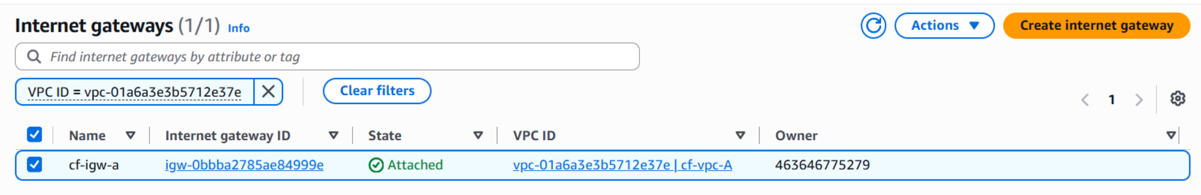
- Create 3 VPCs (same Region, e.g., us-east-1): **cf-vpc-A** (10.0.0.0/16), **cf-vpc-B** (10.1.0.0/16), **cf-vpc-C** (10.2.0.0/16).



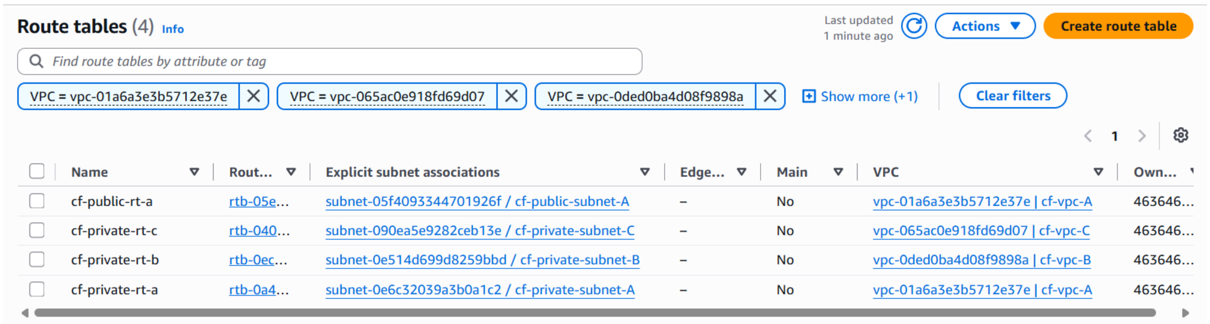
- Create Subnets: **cf-vpc-A** → 1 Public (10.0.1.0/24) + 1 Private (10.0.2.0/24); **cf-vpc-**B → 1 Private (10.1.1.0/24); **cf-vpc-C**→ 1 Private (10.2.1.0/24).



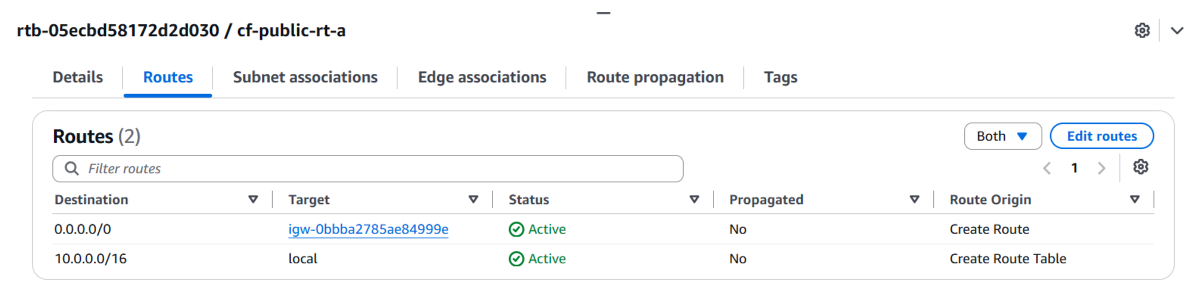
- Create an Internet Gateway (IGW): Attach it only to **cf-vpc-A**.



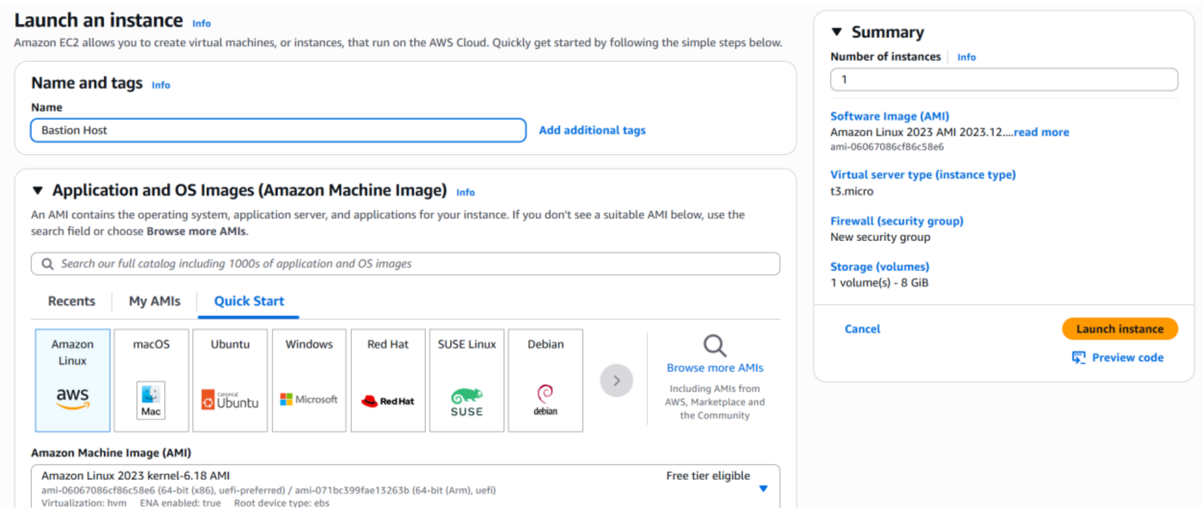
- Create Route Tables: Public RT (**cf-vpc-A**) → 0.0.0.0/0 → IGW; Private RTs → only local route initially.

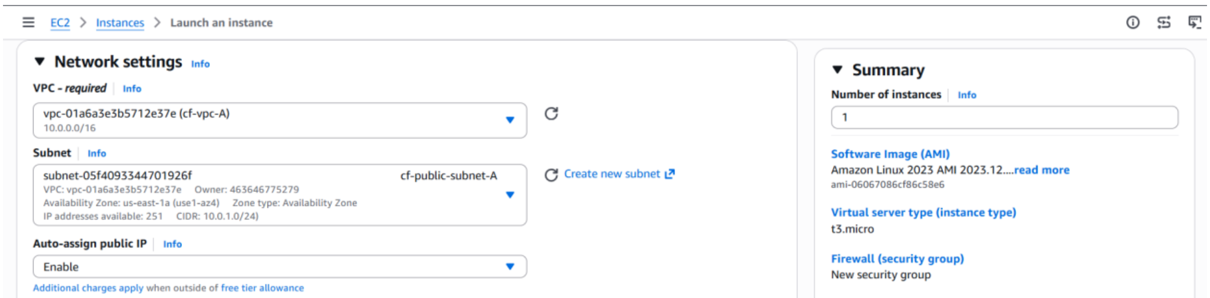


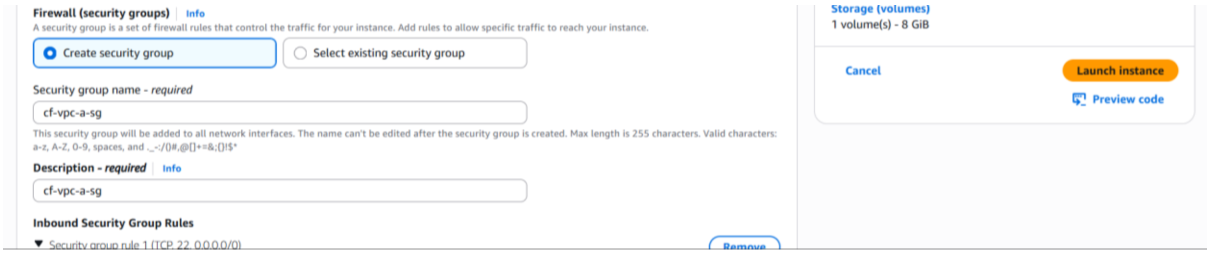
- Routes of cf-public-rt-a:



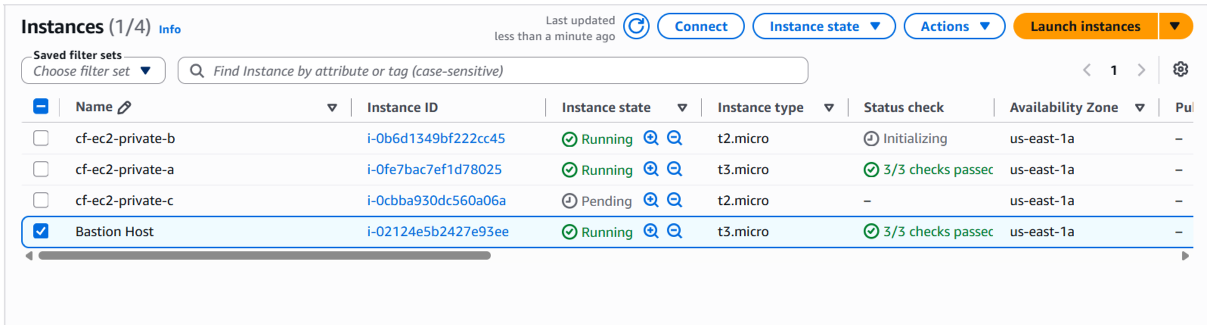
- Launch a total of **four EC2 instances**: one **Jump Host (Bastion Host)** in the **public subnet of cf-vpc-A** for SSH access



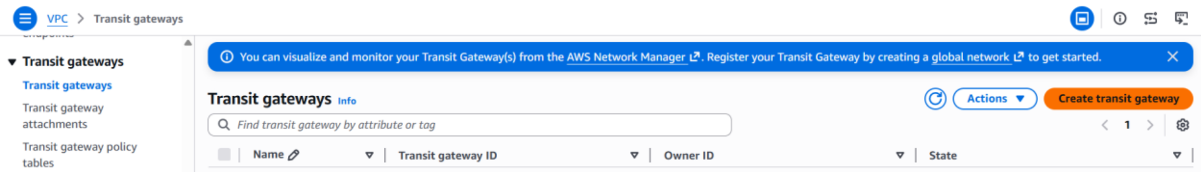




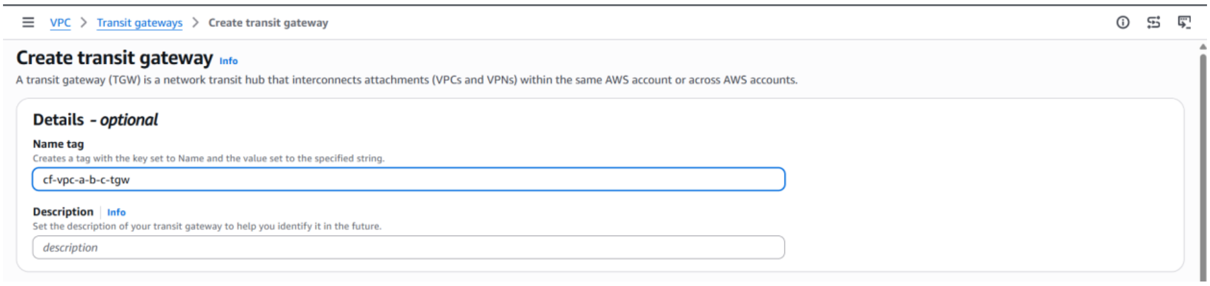
- three **private EC2 instances**—one each in the **private subnets of cf-vpc-A, cf-vpc-B, and cf-vpc-C**—to test connectivity through the Transit Gateway.



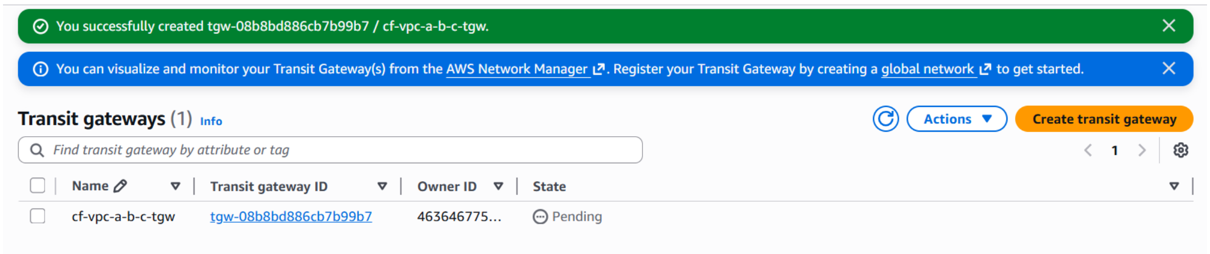
- Navigate to **Transit Gateways** in the VPC console.
- Click **Create Transit Gateway**.



- Provide a meaningful name such as **cf-vpc-a-b-c**.

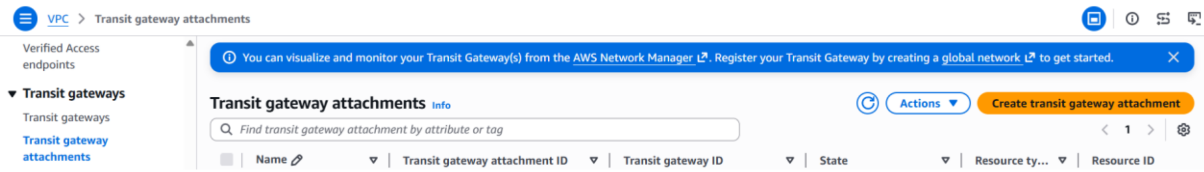


- Leave all other settings as their default values.
- Click **Create Transit Gateway** and wait until its status changes to **Available**.

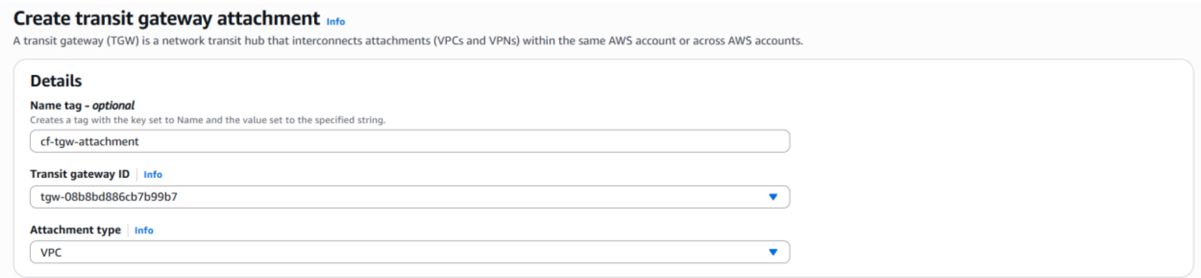


- Open the **Transit Gateway Attachments** page.

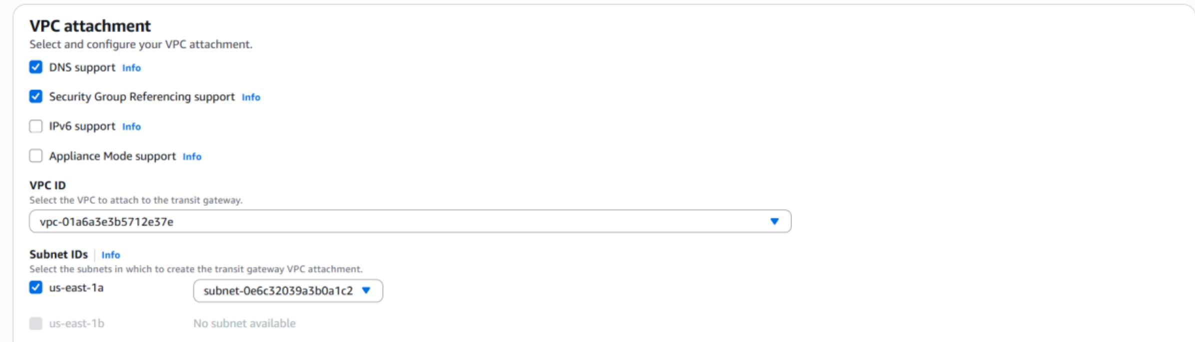
- Click **Create Transit Gateway Attachment**.



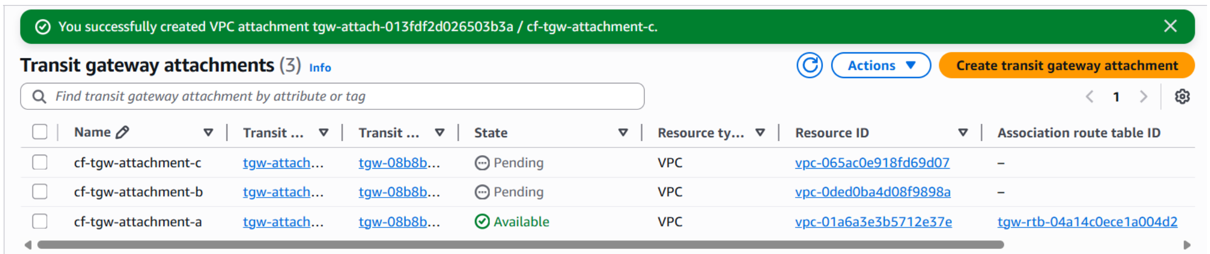
- Select the Transit Gateway that was created in the previous step.
- Choose **VPC** as the attachment type.



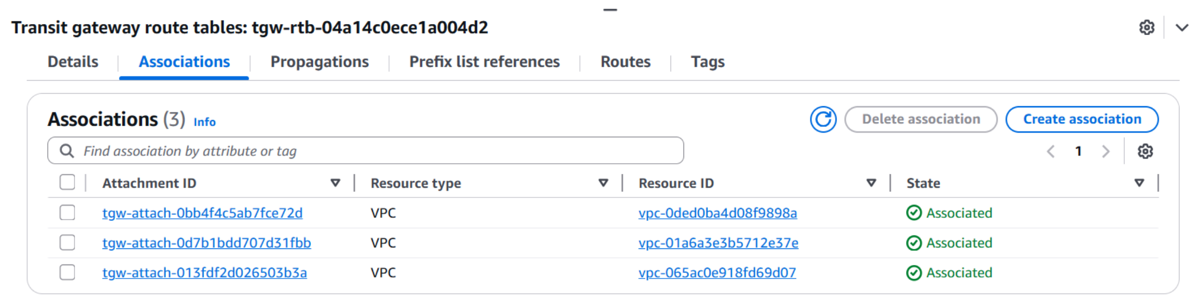
- Select **cf-vpc-A**.
- Choose the **Private Subnet** of **cf-vpc-A** for the attachment.
- Click **Create Attachment**.



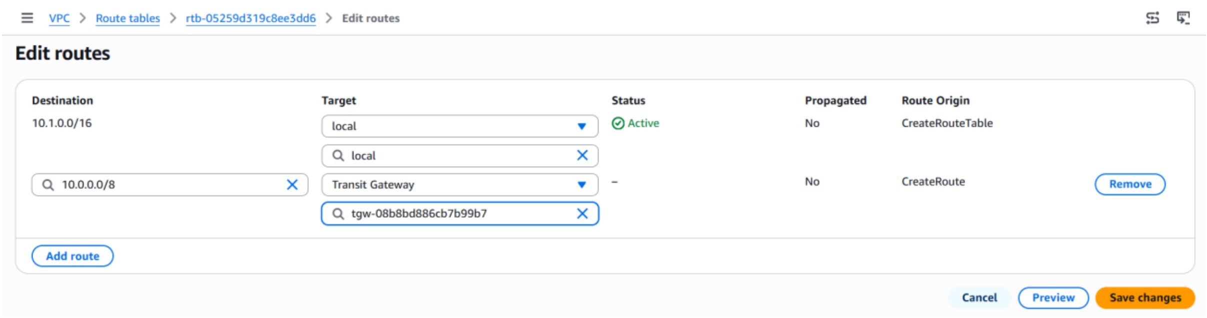
- Repeat the same process for **cf-vpc-B** and **cf-vpc-C**.



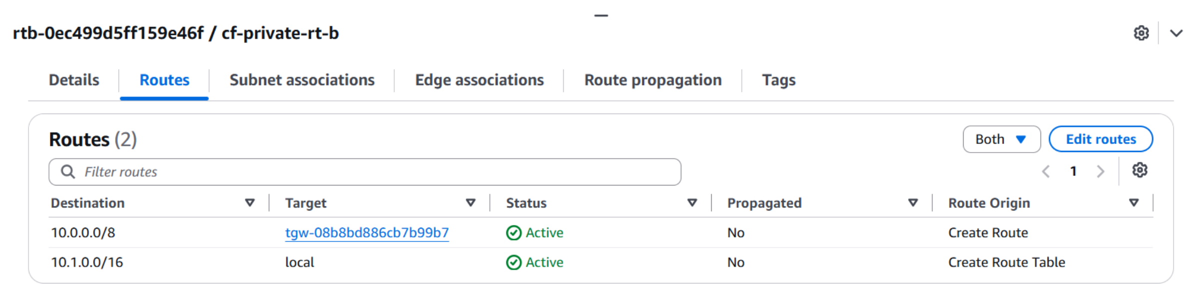
- Open the **Transit Gateway Route Tables** section.
- Select the default route table that was automatically created with the Transit Gateway.
- Open the **Associations**, **Propagations**, and **Routes** tabs.
- Verify that all three VPC attachments are automatically associated and that the CIDR blocks of all three VPCs have been propagated into the Transit Gateway Route Table.



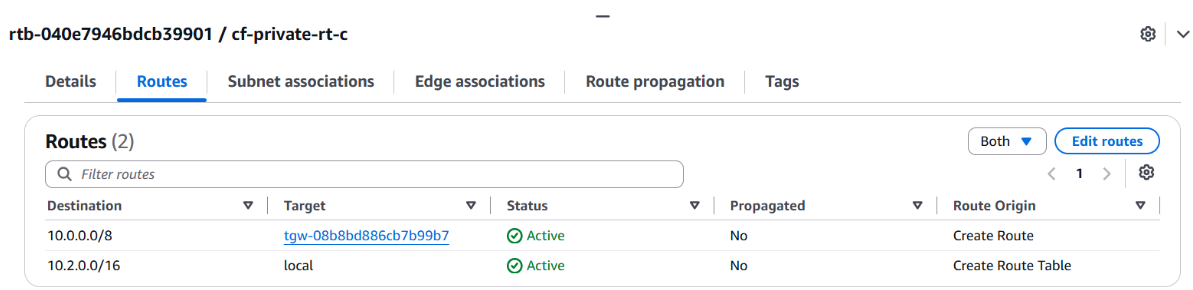
- Open the **Route Table** associated with the **Private Subnet** of **cf-vpc-A, cf-vpc-B** and **cf-vpc-C**.
- Click **Edit Routes**.
- Add a new route with the destination **10.0.0.0/8**.
- Select the **Transit Gateway** as the target.
- Save the route.



- Same for cf-private-rt-b



- And cf-private-rt-c



Do the same thing for the **cf-public-rt-a** rtb-05ecbd58172d2d030 / cf-public-rt-a

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (3)

Both | Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0bbba2785ae84999e	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table
10.0.0.0/8	tgw-08b8bd886cb7b99b7	Active	No	Create Route

Let’s check the traffic for all the instance

This is the traffic for **cf-ec2-private-a, cf-ec2-private-b** and **cf-ec2-private-c**

```
[ec2-user@ip-10-0-1-160 ~]$ ping 10.0.2.254
PING 10.0.2.254 (10.0.2.254) 56(84) bytes of data.
64 bytes from 10.0.2.254: icmp_seq=1 ttl=127 time=0.164 ms
64 bytes from 10.0.2.254: icmp_seq=2 ttl=127 time=0.203 ms
64 bytes from 10.0.2.254: icmp_seq=3 ttl=127 time=0.176 ms
^C
--- 10.0.2.254 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2095ms
rtt min/avg/max/mdev = 0.164/0.181/0.203/0.016 ms
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$ ping 10.1.1.205
PING 10.1.1.205 (10.1.1.205) 56(84) bytes of data.
64 bytes from 10.1.1.205: icmp_seq=1 ttl=126 time=3.05 ms
64 bytes from 10.1.1.205: icmp_seq=2 ttl=126 time=0.849 ms
64 bytes from 10.1.1.205: icmp_seq=3 ttl=126 time=0.841 ms
^C
--- 10.1.1.205 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2059ms
rtt min/avg/max/mdev = 0.841/1.578/3.045/1.037 ms
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$ ping 10.2.1.252
PING 10.2.1.252 (10.2.1.252) 56(84) bytes of data.
64 bytes from 10.2.1.252: icmp_seq=1 ttl=126 time=1.17 ms
64 bytes from 10.2.1.252: icmp_seq=2 ttl=126 time=0.696 ms
64 bytes from 10.2.1.252: icmp_seq=3 ttl=126 time=0.688 ms
64 bytes from 10.2.1.252: icmp_seq=4 ttl=126 time=0.782 ms
64 bytes from 10.2.1.252: icmp_seq=5 ttl=126 time=0.666 ms
^C
--- 10.2.1.252 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4127ms
rtt min/avg/max/mdev = 0.666/0.799/1.165/0.187 ms
[ec2-user@ip-10-0-1-160 ~]$
```